

Shreya S N

Kasaragod, Kerala

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PROFESSIONAL SUMMARY

Final-year Electronics and Communication Engineering undergraduate with a deep interest in core electronics, embedded systems, aerospace systems, and advanced communication technologies. Actively involved in IEEE student activities with experience in leadership, technical coordination, and project development. Passionate about leveraging engineering solutions to contribute to innovation in electronics, embedded systems, communication technologies, and intelligent systems.

EDUCATION

- Govt. Engineering College Kozhikode (APJ Abdul Kalam Technological University, Kerala)
 - BTech.(Hons.) in Electronics & Communication | Minor: Computer Science & Design
 - CGPA: 8.59 | 2023 – Present
- Indian Institute of Technology Madras, Chennai, Tamil Nadu
 - BS in Data Science and Applications (Pursuing Dual Diplomas in Data Science & Programming)
 - CGPA : 7.26 | 2024 – Present

SKILLS & KNOWLEDGE AREAS

- Technical Skills:
 - Programming: Python, C/C++, Embedded C, JavaScript, Verilog
 - Development: React, Flask, HTML, CSS, SQLite
 - Embedded & Hardware: Arduino, ESP32, FPGA, Microcontrollers, IoT, LoRaWAN
 - AI & Data Science: Machine Learning, Deep Learning, Generative AI, Business Analytics
 - Tools & Platforms: MATLAB, LTSpice, Proteus, Git, Linux, VS Code, Google Colab

EXPERIENCE

- ECE Intern at Keltron KKC Kozhikode Jul-2025
 - Worked on embedded systems and Python-based applications using PIC microcontrollers and Arduino platforms. Designed, tested, and validated embedded interfaces and hardware modules for real-world applications, gaining hands-on exposure to embedded development workflows, hardware integration, and system testing.

PROJECTS

- Soil Doctor - KTU Miniproject ESP32, C++, RS485, Embedded C
 - Designed and developed a portable ESP32-based IoT soil monitoring system that measures real-time NPK and soil parameters via an RS485 multi-parameter sensor, delivering intelligent crop and fertilizer recommendations through an LCD interface.

- Reduced manual soil assessment effort by approximately 50% through automated sensing and analysis, eliminating reliance on external lab testing for routine evaluations.
- **Adaptive Wireless Communication Simulator** *MATLAB, App Designer, Communications Toolbox*
 - Built a real-time MATLAB App Designer simulator implementing threshold-based adaptive modulation with dynamic BPSK/QPSK switching over AWGN and Rayleigh fading channels, validated against Proakis & Salehi's *Digital Communications*.
 - Rendered four live diagnostic axes per simulation run—BER vs Eb/N0, IQ constellation, carrier waveform, and eye diagram—alongside a live metrics dashboard tracking BER, throughput, and spectral efficiency.
- **Face Recognition using Deep Learning** *Python, TensorFlow, CNN*
 - Implemented a multi-class face recognition system using a custom CNN architecture (Conv2D, MaxPooling, Dropout, Softmax) trained on the LFW dataset across 60+ identity classes and 3000+ images.
 - Applied data augmentation (flip, rotation, zoom) to address limited per-class samples; evaluated model performance using confusion matrix analysis and training/validation loss curves.
- **RecruitEase - IITM MAD-1 Proj** *Flask, Python, SQLite, HTML*
 - Developed a Flask-based campus placement portal with role-based access control for Admin, Company, and Student personas, each with distinct dashboards and approval workflows.
 - Designed the backend database schema in SQLite to manage user authentication, drive listings, and application state across the full recruitment lifecycle.

LEADERSHIP & ACTIVITIES

- **WIA Representative, IEEE AESS Kerala Chapter** *SLT'26*
 - Contributed to aerospace programs and community engagement for the Women in Aerospace initiative.
- **Proposal Team Lead, IEEE SB GECK** *ExeCom'26*
 - Coordinated the drafting, review, and execution of technical activity proposals.
- **IEEE Day Global Ambassador** *2025*
 - Chosen as one of 32 student ambassadors from the IEEE Kerala Section.
- **Secretary, IEEE WIE Affinity Group, IEEE SB GECK** *ExeCom'26*
 - Directed WIE technical programs and student professional development initiatives.
- **Coordinator, IEEE SIGHT Project (L2L)** *2025*
 - Coordinated a 2 Lakhs IEEE-funded community project from planning to field implementation.
- **NCC A Certificate holder (Naval Wing)** *2017-2019*

LANGUAGES

- Malayalam
- English
- Hindi